

Def: let V be a vector space over F
Any vector of the form $a_1x_1 + a_2x_2 + \dots + a_nx_n$
is called a linear combination of $\{x_i\}$

if $\{x_i\} = \{(0,1), (0,6)\}$ this doesn't suffice.

(a,b) is not always expressible as a linear combination
of $\{(0,1), (0,6)\}$

if $\{x_i\} = \{(1,1), (1,-1)\}$ we saw this is enough
 (a,b) is always expressible as a linear combination
of $\{(1,1), (1,-1)\}$

if $\{x_i\} = \{(1,2), (1,6), (1,-3)\}$ this is enough and
 (a,b) is expressible as a linear combination of $\{(1,2), (1,6), (1,-3)\}$
(infinite possibilities)

Def: if $x_1, x_2, \dots, x_n \in V$ V is a vector space, if y
is expressible as a linear combination of $\{x_1, x_2, \dots, x_n\}$

we say y is in the span of $\{x_1, \dots, x_n\}$

we also write $y \in \text{span}(\{x_1, \dots, x_n\})$

$$(1,2) \in \text{span}(\{(1,1), (1,-1)\})$$

$$\exists a, b \in \mathbb{R} \text{ s.t. } (1,2) = a(1,1) + b(1,-1)$$

Thm: if $L: A \rightarrow B$ is linear and we know the value of L on $\{x_1, \dots, x_n\}$ then we know L on $\text{Span}(\{x_i\})$

PF: Let $w \in \text{Span}(\{x_i\})$ by the definition of Span ,
 $\exists a_1, \dots, a_n$ s.t.

$$w = a_1 x_1 + \dots + a_n x_n$$

$$L(w) = L(a_1 x_1 + \dots + a_n x_n)$$

$$L(w) = a_1 L(x_1) + a_2 L(x_2) + \dots + a_n L(x_n)$$

$\text{Span}\{(0,1), (0,6)\} \neq \mathbb{R}^2$	$\text{Span}\{(1,1), (1,-1)\} = \mathbb{R}^2$
$\parallel$	
$\{(0,a) \mid a \in \mathbb{R}\}$	$\text{Span}\{(1,2), (1,6), (1,-3)\} = \mathbb{R}^2$

Def: if V is a vector space and $x_1, \dots, x_n \in V$ then we say that $\{x_1, \dots, x_n\}$ is linearly

Independent if no $x_i \in \text{Span}\{x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n\}$

no element of $\{x_1, \dots, x_n\}$ can be expressed as a linear combination of the other ones

Q: is $\{(1,2), (1,6), (1,-3)\}$ linearly independent? is there $a, b \in \mathbb{R}$ s.t. $(1,2) = a(1,6) + b(1,-3)$

$$1 = a + b \quad a = 1 - b$$

$$2 = 6a - 3b$$

$$2 = 6 - 6b - 3b \quad -4 = -9b \quad b = \frac{4}{9}$$

So $\{(1,2), (1,6), (1,-3)\}$ is not linearly dependent $a = \frac{5}{9}$